through a number of typical reactions characteristic of strontium. If the radioactivity follows the expected chemistry, then the experimenter can be rather certain that it is strontium. This chemical check does not preclude the possibility that there might be several isotopic nuclides in the sample. However, the presence of several isotopes need not be a handicap in tracer studies concerned with the characteristic properties of a particular element.

When a compound labelled with a radioactive nuclide is used to study molecular behavior, it is necessary to ascertain the *radiochemical purity of the compound* and not just the purity of the radioactive isotope. This means it is necessary to be certain that the radioactivity comes only from a specific element and only from a specific compound incorporating that element. This can be determined through selective chemical operations in which it is frequently desirable that the compound should not be destroyed. Examples of such methods involve preparative gas chromatography, thin-layer chromatography, paper chromatography, dialysis, ion exchange, and solvent extraction. Checking that an element or compound is radiochemically pure is important not only for sources of samples prepared in the laboratory but for commercial samples as well. For example, ^{14}C-tagged organic compounds obtained commercially may have been prepared at varying times prior to purchase. Frequently, the decomposition of molecules in the sample due to the radiolysis that accompanies the radioactive decay results in the presence of products that interfere with the course of the experiment for which the labelled compound is to be used.

In some cases the radioactive product after a period of storage will have grown a sufficient amount of radioactive daughter that it is no longer sufficiently pure radiochemically. In these cases, the daughter activity must be separated prior to use. For example, the ^{90}Y normally present in a ^{90}Sr sample must be eliminated if the tracer solution is to be used to study strontium chemistry.

17.6. RECOIL SEPARATIONS

17.6.1. Target recoil products

When a projectile of high energy reacts with a target atom, the reaction products are emitted in the direction of the projectile, measured in laboratory coordinates. We use the term *recoil* to describe this. The products, whose kinetic energy can be calculated by the conservation laws of Ch. 10, are slowed and eventually stopped through collision with other atoms. Consequently, if the target is relatively thin, there is a certain probability that the recoil range of the product exceeds the thickness of the target and the target atoms can therefore escape the target. In such cases the recoiling products be caught on special foils behind the target (*catcher foils*). These catcher foils can be stationary or they can be a moving band or metal or plastic, or even a gas, which sweeps past the target material.

The recoil technique was used in the discovery of the first transuranium element, neptunium, by E. McMillan and P. H. Abelson in 1939. A uranium salt, $(NH_4)_2U_2O_7$,

was fixed on a paper and surrounded by more layers of thin paper to form a "sandwich" which was irradiated by thermal neutrons. At the time of fission the fission products ejected with high kinetic energy and were caught in the layers of paper. From the number of layers of paper penetrated by individual fission products, their energy and mass could be calculated. A strong β^- activity was found in the first layers of paper, showing that it had relatively little recoil energy. From chemical experiments it was concluded that this activity was not a fission product but was due to ^{239}U which decayed into ^{239}Np:

$$^{238}U(n,\gamma)^{239}U \xrightarrow{\beta^-} {}^{239}Np \tag{17.54}$$

In this particular case, the thermal neutron did not carry any kinetic energy, but the emission of the γ-ray caused the ^{239}U to recoil from the uranium sample prior to decay (with very little recoil) to ^{239}Np. This type of reaction is common in neutron irradiation processes, and is referred to as n,γ-recoil.

Recoil techniques have been used extensively in the synthesis of higher transuranium elements. As an example, consider the formation of element 103, lawrencium (Fig. 17.3). The $_{103}Lr$ nuclides, which were formed in the reaction in the target between ^{252}Cf and projectiles of ^{11}B, recoiled out of the thin target into the helium gas where they were stopped by atomic collisions with the He-atoms. These recoil species became electrically charged cations as they lost electrons in the collision with the gas atoms. They could, therefore, be attracted to a moving, negatively charged, metal coated plastic band, which carried the Lr atom on its surface to a number of α-particle detectors where the α-decay of the lawrencium was measured. From the velocity of movement of the plastic band and the measurement of the radioactivity by energy sensitive nuclear detectors, the half-lives and α-energies of the product nuclei could be determined. Comparison of these data with predictions from theory allowed assignment of the mass number of the nuclide.

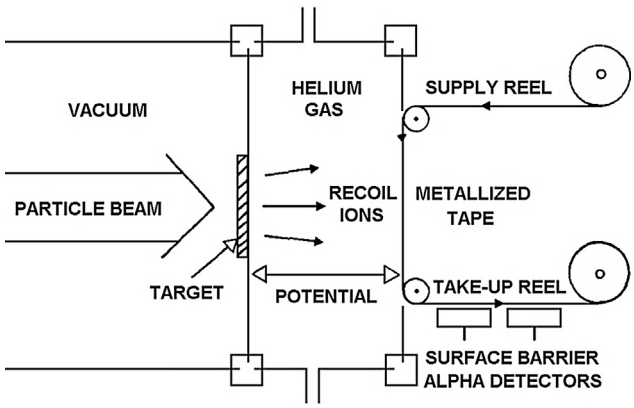


Figure 17.3 Principle of target cage system for α-recoil collection of the products formed.

Target recoil products may have a detrimental effect also. In reactor irradiation of biological samples to determine their trace metal concentration it was found that n,γ products formed in the material of the sample container recoiled out of the container wall into the sample, contaminating it. Such contamination can be eliminated by using a container which forms a minimal amount of such products.

17.6.2. Hot atom reactions

The momentum imparted to a nucleus in a nuclear reaction with a charged particle or a fast neutron almost invariably is sufficient to result in the rupture of chemical bonds holding the atom in the molecule. The same is true for the recoil energy imparted to a nucleus by the emission of an α-particle as well as in β-decay, γ-decay, etc.; the recoil energies in radioactive decay are discussed in §§5.3–5.5.

These recoiling atoms are known as *hot atoms* since their kinetic energy is much in excess of thermal equilibrium values. A hot atom may move as much as several hundred atomic diameters after the rupture of the bond before being stopped even though this takes only about 10^{-10} s. Initially the hot atom is highly ionized (e.g. for $Z = 50$ ionic charges up to $+20$ have been observed). As the hot atom is slowed down to thermal energies, it collides with a number of other particles in its path producing radicals, ions, and excited molecules and atoms. At thermal energies it may become a neutral atom (particularly if the matrix material is metallic), an ion, possibly in a different oxidation state (a common situation in inorganic material), or it may react to form a compound with molecules in its path (the prevalent situation in matrices of covalent material).

Chemical effects of nuclear transformations (*hot atom chemistry*) have been extensively studied in connection with induced nuclear transformations, both in the gas phase, in solution and in the solid state. In the latter cases the dissipation of the kinetic energy and neutralization of the charge within a small volume produces a high concentration of radicals, ions, and excited molecules in the region where the recoiling atom is slowed to energies where it can form stable bonds. Usually the product molecule is labeled neither very specifically nor completely randomly. The topic of hot atom behavior is also treated in Ch. 8.

Hot atom reactions can be used for synthesis of labelled compounds. So far this technique has been used primarily for ^{3}H and ^{14}C labelling. For example, an organic gaseous compound is mixed with ^{3}He and the sample is irradiated with neutrons. The reaction $^{3}He(n,p)^{3}H^{\rightarrow}$ produces tritium as hot atoms (indicated by the small arrow) with kinetic energies of approximately 0.2 MeV. These hot atoms react with the organic molecules to produce labelled compounds. An example is the formation of labelled butanol

$$C_4H_9OH + {}^{3}H^{\rightarrow} \rightarrow C_4H_9O{}^{3}H + {}^{1}H^{\rightarrow} \tag{17.55}$$

The reaction is referred to as a *hot atom displacement* reaction. In an alternate method the solid organic compound may be mixed with Li_2CO_3, and this solid mixture irradiated with neutrons to cause the reaction: $^{6}Li(n,\alpha)^{3}H^{\rightarrow}$. Again the hot tritium atom combines with the organic species to cause labeling. In a third and simpler method, the organic compound is simply mixed with tritiated gas. The β-radiation from tritium decay causes excitation and decomposition which leads to capture by the organic radicals of some of the remaining tritium. These reactions are not selective because reactions with decomposition products also occur. Therefore the tritium labeled product must be purified. Gas phase labeling usually results in incorporation of tritium in several sites of the molecule, e.g. toluene has about 9% of the total label in the CH_3 group, 54% in the *ortho* sites of the ring, 24% in the *meta* sites, and the remaining 13% at the *para* site.

^{14}C-labeling can be achieved in a similar manner by using the $^{14}N(n,p)^{14}C^{\rightarrow}$ reaction. The recoiling ^{14}C energy is about 0.6 MeV. A suitable labeling mixture is NH_4NO_3 + the organic compound: cf. ref. Wolf (1960).

17.6.3. The Szilard-Chalmers process

In 1934 L. Szilard and T. A. Chalmers discovered that bond breaking could occur for atoms following nuclear reaction or radioactive decay even though the recoil energy in the initial process is not sufficient to overcome the bonding energy. In the case of thermal neutron capture the processes involved in the emission of the γ-ray, which removes the nuclear excitation energy, impart recoil energy to the atom to break most chemical bonds (n,γ-recoil). If, after rupture of the bonds, the product atoms exist in a chemical state different and separable from that of the target atoms, the former may be isolated from the large mass of inactive target. This provides a means of obtaining high specific activities in reactions where target and product are isotopic.

This process is known as the *Szilard-Chalmers reaction* and was discovered when, following the irradiation of ethyl iodide with thermal neutrons, it was found that radioactive iodide could be extracted from the ethyl iodide with water. Moreover, when iodide carrier and silver ions were added to this aqueous phase, the radioactive iodide precipitated as silver iodide. The obvious interpretation of these results is that the neutron irradiation of the ethyl iodide, which caused the formation of ^{128}I, ruptured the bonding of this atom to the ethyl group. The bond energy of iodine to carbon in C_2H_5I is about 2 eV. Since this exceeds the recoil energies of neutron capture, the bond breakage must have resulted from the γ-emission which followed neutron capture and not the capture process itself. The reaction can be written:

$$C_2H_5\,{}^{127}I + n \;\rightarrow\; [C_2H_5\,{}^{128}I]^{*} \;\rightarrow\; C_2H_5 + {}^{128}I^{\rightarrow} + \gamma \qquad (17.56)$$

The $^{128}I^{\rightarrow}$ loses its kinetic energy and is stabilized as an iodine atom or iodide ion; it can also be recaptured by the C_2H_5 radical (*retention* of activity in C_2H_5I). In addition to

the necessity that the recoiling species have sufficient energy to rupture the bond, it is also necessary for a successful enrichment of specific activity by the Szilard-Chalmers process that there is no rapid exchange at thermal energies between the active and inactive iodine atoms in ethyl iodide:

$$C_2H_5{}^{127}I + {}^{128}I \overset{\text{slow}}{\rightleftharpoons} C_2H_5{}^{128}I + {}^{127}I \tag{17.57}$$

After the rupture of the bond, the hot atoms may also replace iodine or even hydrogen atoms from ethyl iodide molecules, similar to the thermal exchange:

$${}^{128}I^{\rightarrow} + C_2H_5{}^{127}I \rightarrow C_2H_5{}^{128}I + {}^{127}I \tag{17.58}$$

Again, the result is retention of the activity within the organic molecule. Retention is decreased by diluting the ethyl iodide with alcohol which reduces the probability of collision and exchange with C_2H_5I molecules as the hot atoms of ${}^{128}I^{\rightarrow}$ are being slowed.

Isomeric transitions which proceed by emission of γ-rays may not provide sufficient recoil energy to break covalent bonds. However, in these cases or very low energy isomeric transitions, the extent of internal conversion is large. This results in vacancies in the inner electron orbitals (§5.5). When electrons in outer orbitals move to fill the vacancies, the difference in electron binding energies is sufficient to cause some ionization, resulting in relatively high charge states for the atom, leading to bond rupture. An example of such a process is seen in the reaction whereby an alkyl bromide is irradiated with thermal neutrons. During neutron irradiation the reactions are

$$\text{RBr} + \text{n} \begin{cases} \xrightarrow[\sigma\ 2.9\ \text{b}]{} \text{R} + {}^{80\text{m}}\text{Br}\ (4.4\ \text{h}) \\ \qquad\qquad \downarrow \text{IT} \\ \xrightarrow[\sigma\ 8.5\ \text{b}]{} \text{R} + {}^{80}\text{Br}\ (18\ \text{min}) \end{cases} \tag{17.59}$$

Extraction by water of this bromide produces an aqueous sample containing both ${}^{80}Br$ and ${}^{80m}Br$. Some ${}^{80}Br$ and ${}^{80m}Br$ are retained as alkyl bromide in the organic phase. If, after a period of an hour, a new extraction is carried out on the organic phase, only the ${}^{80}Br$ is found in the aqueous sample. This ${}^{80}Br$ results from the following reaction

$$R{}^{80m}Br \xrightarrow{\text{IT}} R + {}^{80}Br + \gamma \tag{17.60}$$

In addition to the organic halides a number of inorganic systems undergo similar reactions. From neutron irradiation of acid or neutral solutions of permanganate most of the ${}^{56}Mn$ activity is removable as MnO_2. From solid or dissolved chlorates, bromate, iodate, perchlorates, and periodates, active halide samples may be isolated in relatively high yields. Systems with Te, Se, As, Cu, and several Group VIII elements have been shown to be applicable to the Szilard-Chalmers process. The great advantage in all these

cases is that a relatively small amount of activity may be isolated from a large mass of isotopic inactive target. Because of retention there is also some labeling of the target molecules.

17.7. FAST RADIOCHEMICAL SEPARATIONS

Short-lived radioactive nuclides are of interest (i) to nuclear science to establish the outer boundary for nuclide formation, which yields information on nuclear structure , (ii) to chemistry to extend the limits of the periodic system, i.e. the actinides and transactinides (or *superheavy elements*); these studies are also of interest to cosmology since these heaviest elements are believed to be formed in supernovae, and (iii) to medicine where they offer a unique tool to identify and treat some diseases (§18.5). Short-lived radionuclides are also used in chemical and industrial research, one of their advantages being that they rapidly disappear, eliminating radioactive waste problems. We arbitrarily define short-lived as meaning half-lives <10 minutes. Because many of the lightest elements of greatest interest in organic chemistry and biology lack long-lived radionuclides, the production of their short-lived isotopes are of particular interest; see Figure 5.7. At the other end of the periodic system there are no stable isotopes at all, and for the trans-uranium elements the half-lives of the most stable isotopes become shorter and shorter for each higher element, e.g. the longest-lived isotope of $Z = 105$ has $t_{½} \sim 34$ s. Thus, research on the production of short-lived radionuclides has been particularly intense for the lightest and heaviest elements as well as for the fission products . It is obvious that studies and uses of these short-lived nuclides require special techniques, which often are rather costly. A brief description of some of the techniques used follows.

The sequence of procedures necessary for the production and study of very short-lived nuclei are:

1) target preparation and irradiation,
2) primary (preliminary) target purification (at irradiation site),
3) transport of product (e.g. through pneumatic tubes in which the transport velocity may be up to 100 m s^{-1}) from irradiation site to nuclear chemistry laboratory,
4) isolation of desired product and eventual labeling, and
5) final use in physico-chemical experiments or in applied areas; e.g. hospitals or industry.

A primary requirement for work with short-lived nuclides is that the site or production and the site of use be close together. In 2–30 seconds it is possible to use rapid separation procedures consisting of (a) target dissolution, followed by (b) chemical isolation of the radionuclide through solvent-extraction-stripping or high pressure/temperature ion exchange, and/or (c) precipitation or electro-plating before (d) measurement. In this scheme eventual chemical studies can be included, usually between steps (b) and (c).